How to make a BikeHide

Parts

Tarpaulin - a car / truck breaker in RenfRew, NT Technologies, on Meadowside Street, provided Tarpaulin for a small tip.
https://www.google.com/maps/place//@55.8863781,-4.3901176,151m/data=!3m2!1e3!4b1?entry=ttu&g_ep=EgoyMDI2MTEwNC4wIKXMDSoASAFQAw%3D%3D

Seatbelts from the same.

20mm x 3m galvanized pipe - from Screwfix - "Deta TTE Round Steel Galvanised Class 4 Conduit Tube 20mm x 3m" - https://www.screwfix.com/p/deta-tte-round-steel-galvanised-class-4-conduit-tube-20mm-x-3m/7734j (actually 2m would be better, but you can join two 1m ends to make a 3rd pipe)

Eyelets - 6mm brass eyelets - from "The Trimming Shop" in London - https://trimmingshop.co.uk/products/brass-eyelets-100pcs?_pos=1&_sid=02fd45dc3&_ss=r&variant=52528083403091
8mm brass eyelets are useful for seatbelt handles

Bolts - choose stainless steel. All bolts are M5.
30mm M5 roofing bolts - for attaching pipes - https://www.screwfix.com/p/easyfix-bright-zinc-plated-roofing-bolts-m5-x-30mm-10-pack/6194j

M5 12mm bolts - https://www.screwfix.com/p/easyfix-button-head-socket-screws-a2-stainless-steel-m5-x-12mm-50-pack/8341t
16mm are also useful - https://www.screwfix.com/p/easyfix-button-head-socket-screws-a2-stainless-steel-m5-x-16mm-50-pack/8345t

Nuts - M5 stainless steel Nylock - https://www.screwfix.com/p/easyfix-a2-stainless-steel-nylon-lock-nuts-m5-100-pack/1125t

Penny washers for attaching seatbelt handles.

Rope - 4mm or 5mm rope is useful. Choose one with a sheath that is comfortable to touch

Strap for lifting the cover - e.g. climbing tape (or use rope)

Paint - Zinsser AllCoat Exterior Matt - Black - works well for painting tarpaulin. IT lasts on flexing, and copes with weather https://zinsseruk.com/products/allcoat-exterior-matt

Thin waterproof fabric for sides of handlebar pockets.

Flat aluminium strip can be used for reinforcing around handlebar pockets if necessary.

BikeHide stickers.

unnecessary parts

- nylon p-clips are superfluous and slip - better to drill and bolt the pipes

Tools

Eyelet punches ("ZYT pliers") and bits come from the trimming shop - 6mm (and 8mm) dies and hole punches are useful https://trimmingshop.co.uk/products/zyt-hand-press-pliers-with-eyelets-die-tool?_pos=4&_psq=zyt+pliers&_psid=76cfea3a4&_ss=e&variant=52527984017747

It's also useful to have a manual punch for holes far from the edges of tarpaulins (or mount the lower dies (as above) on a piece of wood, and hammer the upper one from above)

Also:

- electric screwdriver
- hammer
- drill + metal bits
- spanner + allen keys
- paint brush & cloth (the water based paint washes quite easily)
- Brushes + hose pipe for cleaning tarp
- strong scissors for cutting tarp
- tape measure, ruler, straight edge, right angle
- lighter for singeing ends of ropes
- hacksaw + file for cutting pipes
- glue and tape can be useful.

Cutting machines would be useful! Laser cutting PVC is possible, but releases Chlorine gas, so needs good ventilation.

Process

Obtain tarpaulin. Strip of metal parts and clean (hose + brush useful). Cut into sections approx 1.9m wide x 2.2m approx high per cover (see diagram). Take care with the layout of the vertical straps that cross most tarps: the cuts should be arranged such that the handlebar pockets lie between these. Excess material can be used for handlebar pockets

Paint the tarpaulin. One coat seems ok. Remember you'll need a strip approx 45cm wide for the handlebar flap extensions.

See the diagram below for measurements, which can be adjust according to the bike, but generally work well.

Mark out lines for the pipes. One on the centreline, then the others 2/3 of the way to the bottom (this proportion makes the cover lift most efficiently). One side of the cover can be cut longer than the other (since the bike takes up space on one side more than the other), or you can use an extension hem, or make long and allow cutting later, as you prefer.

Mark holes for eyelets, and holes and same points on pipes. Use double holes approx 1 or 2 inches apart on the ends (these spread load and resist wear better, where the tarp flexes). Double holes in the middle of the centre pipe can allow ropes to replace bolts, to allow the cover to be suspended from above by the centre pipe.

Cut handlebar flaps in tarpaulin, with care (three cuts, in a square cup shape, with the top edge, closest to middle pipe, still attached).

Drill pipes, and attach to eyeletted holes with 30mm roof bolts. Generally it's best to put poles made from joining two 1m ends on the middle pole, as it moves less.
Make handlebar flap extensions. 14mm gaps on a square arrangement works well. A stencil for laying out the eyelet holes helps. Take care! Regular gaps are best.

Punch holes and add eyelets around edge and bottoms of handlebar flaps (tedious). Make the extensions have a square eyelet hole arrangement if possible, for maximum flexibility.

Cut and attach fabric side panels. A rectangle with two corners removed (triangles chopped off) works reasonably, sized a little larger than the edges of the handlebar flap (including the extension at its longest). Put eyelets on a doubled hem of this, and attach to the inside of the handlebar pocket. Excess best gathered at bottom (multiple eyelets on one bolt), allowing for adjustment. Frequent eyelets and bolts on the curved section of the handlebar flap help resist wear from handlebars, and their escaping out into the open air.

Attach seatbelt handles. 8mm eyelets seem to work better than 6mm, which slip too easily in the fabric. Penny washers help hold it on.

Small sections of rope are useful to attach the hide to fence. You can put holes in the top pipe to allow rope to come out the top too, to allow suspension of the hide (e.g. on a high fence). In this case, a hook is useful to allow the hide to be held open.

A righting strap is useful if the hide is put over a fence, to allow it to be flung over the far side of the fence then pulled back.

"BikeHide" stickers can be put on handlebar pockets. Superglue helps!

You can dangle a bar inside for tools. You can add holes and chains to allow the cover to be held shut. A tie (with carabiner?) from pedal to handle can keep the cover down in strong wind.

Variants

Cover for two bikes - add 15 or 20cm to the width, and possibly double the handlebar flaps (or have a big long flap, with 15 to 20cm margins at ends, and maybe rope or ties to allow adjustment?). This idea from seeing people do it in practice.

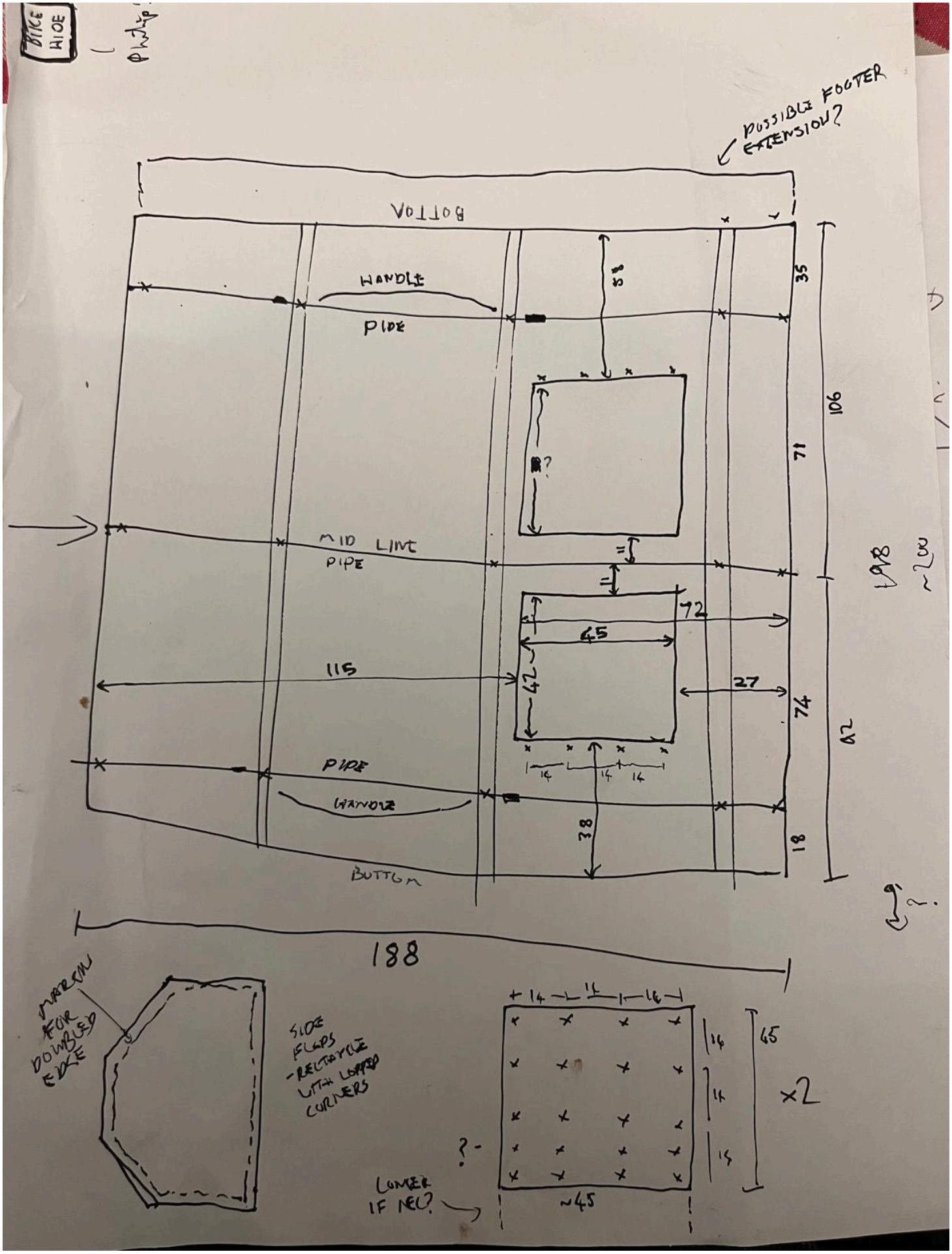
Cut in middle - cut the cover in two, along the middle pole. Add a metal strip on one edge of the cut, allowing the two bits to be joined with bolts. This allows mounting on fence spikes (tested, works), and through high fences.

Optional extension flap of 15-20cm extra on bottom - like hem of skirt. Allows more coverage. Use eyelets to allow attachment / removal / swapping sides.

Small strips of flat aluminium can add structure to the handlebar flaps if necessary.

Chain for security - either lock the pipes at ends with chain and padlocks. Or use solid metal on one pipe (heavy!).

Decorate the outside (graffiti....)


BIKE HOE
POSSIBLE FOOTER EXTENSION?
BOTTOM
HANDLE
PIPE
MID LINE PIPE
PIPE
HANDLE
BOTTOM
35
106
71
74
42
18
38
72
45
42
27
115
188
MARGIN FOR DOUBLED EDGE
SIDE FLAPS - RECTANGLE WITH LOPPED CORNERS
LOWER IF NEC?
~45
65
x2